

Assignment 2

Due Date: November 5, 2010 by 5:30 pm

1. (5 points) Write a python code to multiply two 3 by 3 matrices A and B:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$$

Note: for this question and any other that uses arrays, you will need to import the module “numpy”.

2. (10 points) Using the two 3 by 3 matrices defined in question 1, write a python code to do the long version of the matrix multiply. That is, do not use the dot() function provided in numpy to generate the matrix product.
3. (10 points) Write a python script that takes a number between 1 and 10 as input and writes the word equivalent of the number. In other words, the script should take an integer, say 5, as input and output the word “five” to the screen. Have the script read the value from STDIN.

So, if I run this script using ./numconv.py and hit the return key, it will wait for me to type a number (say I type 5), then return the word equivalent to the screen, in this case:

five

Note: you will need the module sys to read from STDIN.

4. (6 points) Write a python code that stores the physical constants

Avogadro's constant
Planck constant
Proton mass
Rydberg constant
Electron mass
Newtonian constant of gravitation
Speed of light in vacuum
Fine structure constant

And returns the value of the physical constant that is requested as an option to the script. In other words, the script takes the name of a physical constant as input and returns its value. So, if the script is called “phycon.py”, and I run it using ./phycon.py, hit the return key and type proton_mass when prompted, it will give something like:

1.672e-27

Do not worry about units or the precision of the number returned, as long as it is close to the actual physical constant.

5. (5 points) Write a python script to print the numbers of the Fibonacci series below 1000. The Fibonacci series is a set of numbers defined by the recurrence relation (noting that $F_0 = 0$ and $F_1 = 1$):

$$F_n = F_{n-1} + F_{n-2}$$